

Data Newbies Hub Challenge



Analyst

69

Working

37



Tools

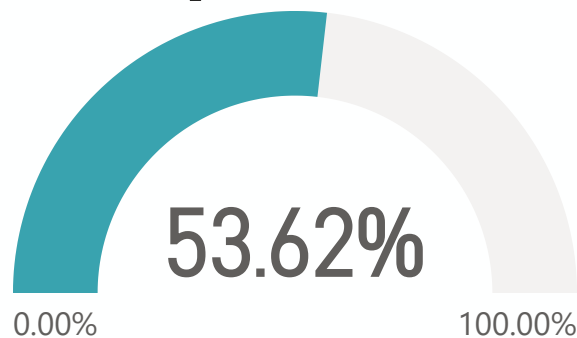
5

Avg tools used

2

Excel 81% and **PowerBI 65%** are the most used tool, **SQL 46%** and **Python 28%** come 3rd and 4th, with **R 14%**

Total Working

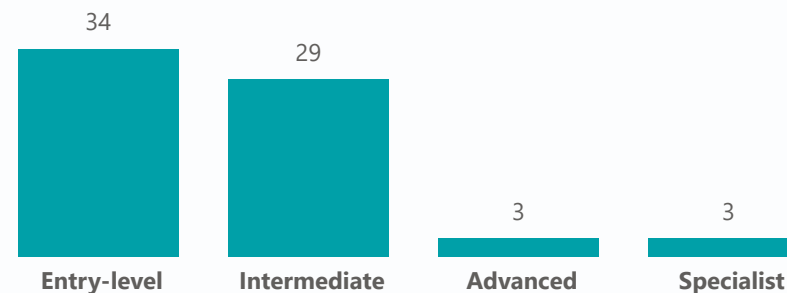


Not Working

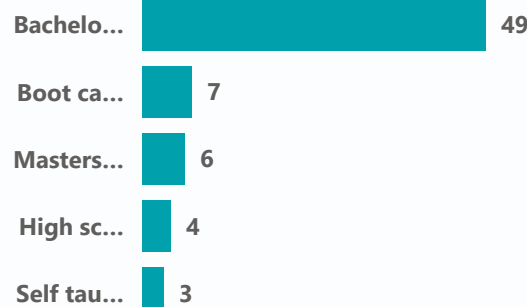
Working

Satisfaction_clean	Respondents
Not Satisfied	30
Somewhat Satisfied	34
Very Satisfied	5

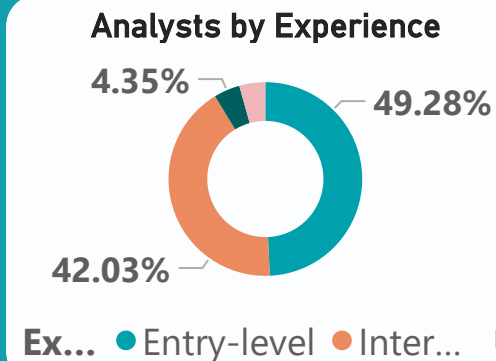
Analyst by Education level



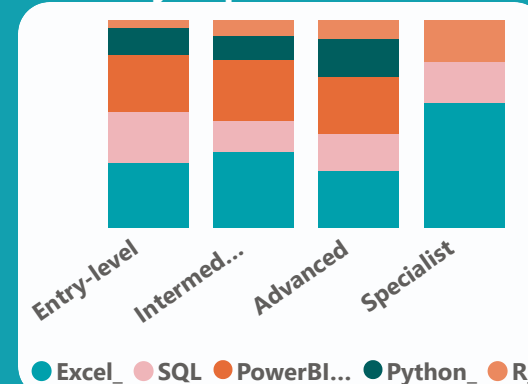
Analyst by Education level



Analyst by Education level



Tools by Experience

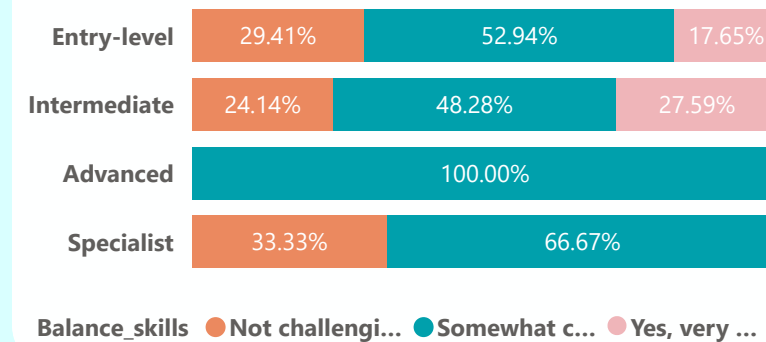


Job Ease by Education

Job_ease ● Difficult ● Somewhat easy ● Very difficult ● Very easy

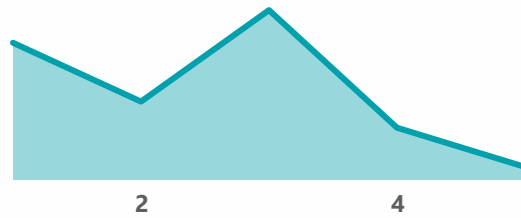


Balancing Skills by Experience_level





Tools Count by Analyst



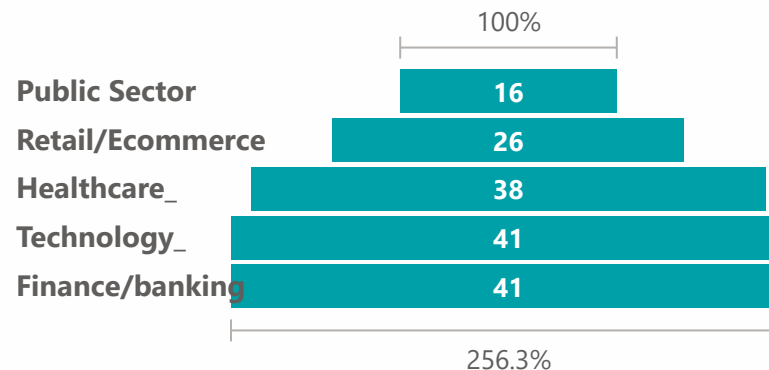
49% of respondents are **Entry-level**, **42%** are **Intermediate**, while **Advanced** and **Specialist** share a tie of **4.3%**. **Excel** is the most used tool in all experience levels. **R** is barely used.

There are only **6 Respondents (8.6%)** who are very satisfied with their roles, all these people use a minimum of **3 tools**, of which use a BI tool (**PowerBI/Tableau**) is constant.

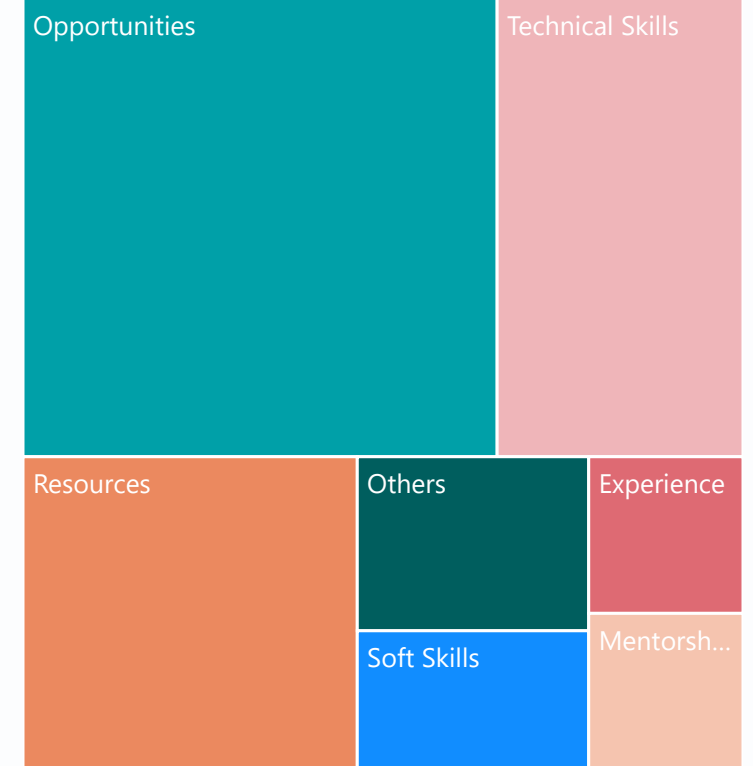
Finance/banking and Technology are **Joint best rated Industries (41)**, **Healthcare(38)** and **Retail/E-commerce(37)** come as **2nd** and **3rd** with **Government/Public (16)** last.

Valuable_skill	Respondents
Business Acumen	5
Communication and storytelling skills	48
Leadership and Management	2
Technical skills	14

Industry by Analyst

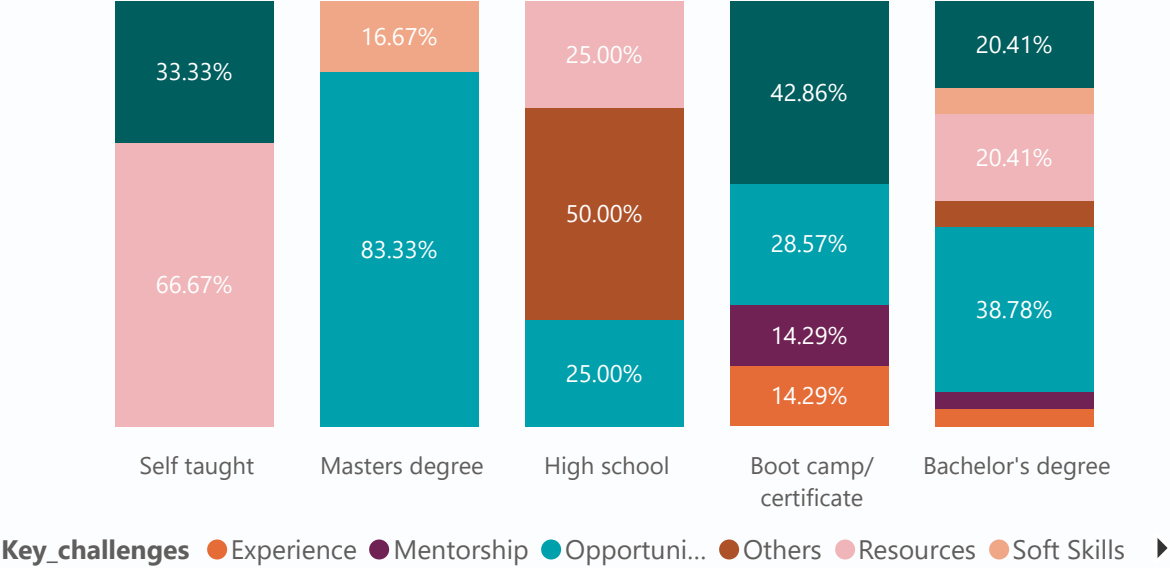


Challenges Faced by Analyst



Satisfaction_clean	Respondents
Not Satisfied	30
Somewhat Satisfied	34
Very Satisfied	5

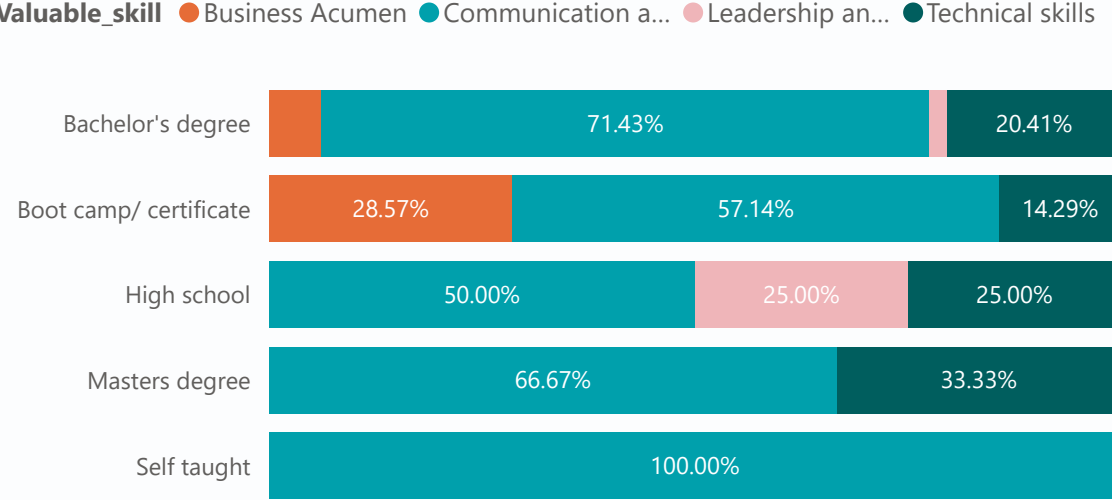
Respondents by Education and Key_challenges



Key_challenges	Finance/ banking	Technology_	Healthcare_	Retail/ Ecomm erce	Public Sector
Opportunities	15	14	17	11	6
Technical Skills	9	11	8	4	4
Resources	9	8	7	4	2
Experience	2	3	2	1	1
Mentorship	3	2	0	1	1
Soft Skills	2	2	2	2	1
Others	1	1	2	3	1

Valuable_skill	Finance/ banking	Technology_	Healthcare_	Retail/ Ecomm erce	Public Sector
Communication and storytelling skills	29	34	29	17	10
Technical skills	8	6	7	5	4
Business Acumen	3	1	2	3	2
Leadership and Management	1	0	0	1	0

Respondents by Education and Valuable_skill



DATANEWBIES HUB PROJECT: Leveraging Career Data for Strategic Decision Making

Title: Challenges and Opportunities in Data Analysis
Career

Date: 18-01-2025

Presenter: Ayobami. O. Oje

OVERVIEW OF THE ANALYSIS

OBJECTIVE:

The goal of this analysis is to examine the challenges and Opportunities in Data Analysis Career, through this we can derive actionable insights that inform better decision-making for aspiring data analysts.

KEY AREAS:

Respondents Demographics: Understanding the distribution of responders by experience level, education level, industry representation etc

Tool Popularity: Understanding the Most Used Tools, percentage of respondents using each tool, tool Usage by Experience Level, comparison of tool usage across different levels of experience, average Number of Tools Used: the mean number of tools mentioned per respondent.

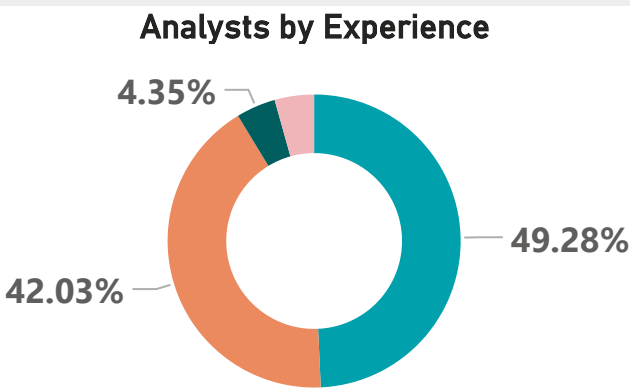
Career and Skills: Understanding reasons respondents entered the data field, common obstacles faced by respondents, such as lack of resources or job opportunities, satisfaction Level: Average satisfaction score for respondents in their current roles or learning paths.

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	ID	Experien	Too	Motiva	Educat	Challen	Job_stru	Balance	Valueable	Indus	Satisfac	Job_e	
2	1	Less than a year	SQL, Po	Good salary	High school	Option 1	I haven't found	Not challenging	Communication	Finance/b	1-3(Not satis	Difficult	
3	2	1 - 3 years	Excel	Opportunitie	Bachelor's	Getting dat	Yes	Somewhat chal	Communication	Finance/b	4-5(somewh	Very difficult	
4	3	1 - 3 years	Excel	Interested in	Bachelor's	Data story t	No	Yes, very chall	Communication	Healthcare	4-5(somewh	Somewhat easy	
5	4	Less than a year	Excel, St	Good salary	Bachelor's	Difficulty in	Yes	Yes, very chall	Communication	Finance/b	4-5(somewh	Very difficult	
6	5	Less than a year	Excel, P	Opportunitie	Boot camp	Data cleanin	I haven't found	Somewhat chal	Communication	Finance/b	7-8(satisfie	Somewhat easy	
7	6	3 - 5 years	Excel, St	Good salary	Boot camp	When stuck	I haven't found	Somewhat chal	Communication	Finance/b	7-8(satisfie	Very difficult	
8	7	1 - 3 years	Excel, P	Opportunitie	Bachelor's	The fast pa	I haven't found	Somewhat chal	Business Acum	Retail/E-c	4-5(somewh	Difficult	
9	8	1 - 3 years	Excel	Interested in	Masters de	I am trying t	No	Somewhat chal	Communication	Technolog	1-3(Not satis	Difficult	
10	9	1 - 3 years	Excel, St	Interested in	Bachelor's	Getting a cl	Yes	Not challenging	Technical skills	Finance/b	4-5(somewh	Difficult	
11	10	Less than a year	Excel, St	Interested in	Bachelor's	Data cleanin	Yes	Yes, very chall	Communication	Technolog	7-8(satisfie	Difficult	
12	11	Less than a year	Excel	Good salary	Masters de	Getting a jo	I haven't found	Somewhat chal	Communication	Healthcare	7-8(satisfie	Somewhat easy	
13	12	Less than a year	Excel, St	Interested in	High school	Getting clie	I haven't found	Not challenging	Communication	Technolog	9-10(very sat	Somewhat easy	
14	13	Less than a year	Python	High demand	Self taught	Difficulty in	Yes	Not challenging	Communication	Technolog	1-3(Not satis	Very difficult	
15	14	1 - 3 years	Excel	Good salary	Masters de	Difficulty in	Yes	Not challenging	Communication	Finance/b	Somewhat sa	Difficult	
16	15	Less than a year	Excel	Opportunitie	Bachelor's	Still trying t	I haven't found	Somewhat chal	Technical skills	Finance/b	Not satisfied	Difficult	
17	16	1 - 3 years	Excel, P	Interested in	Bachelor's	Getting an i	No	Not challenging	Technical skills	Finance/b	Very satisfie	Somewhat easy	
18	17	1 - 3 years	Excel, P	Opportunitie	Self taught	Getting my	I haven't found	Yes, very chall	Communication	Finance/b	Not satisfied	Very difficult	
19	18	1 - 3 years	Excel, P	Interested in	Bachelor's	Learning R	I haven't found	Not challenging	Technical skills	Finance/b	Somewhat sa	Somewhat easy	
20	19	Less than a year	Excel	Opportunitie	Bachelor's	Creating tin	I haven't found	Yes, very chall	Business Acum	Retail/E-c	Somewhat sa	Very difficult	
21	20	5+ years	Excel, St	Good salary	Boot camp	The market	Yes	Not challenging	Business Acum	Finance/b	Satisfied	Difficult	
22	21	1 - 3 years	Excel, St	High demand	Masters de	Difficulty in	No	Somewhat chal	Communication	Finance/b	Very satisfie	Somewhat easy	
23	22	Less than a year	SQL, Po	Opportunitie	Bachelor's	Getting a jo	I haven't found	Not challenging	Communication	Retail/E-c	Satisfied	Very difficult	
24	23	1 - 3 years	Excel, P	Opportunitie	Bachelor's	Accessing t	Yes	Yes, very chall	Communication	Finance/b	Somewhat sa	Difficult	
25	24	1 - 3 years	Excel, P	Opportunitie	Bachelor's	Visualizing	I haven't found	Somewhat chal	Communication	Technolog	Not satisfied	Difficult	
26	25	1 - 3 years	Excel, P	Good salary	Bachelor's	Time factor	I haven't found	Somewhat chal	Communication	Retail/E-c	Not satisfied	Very difficult	
27	26	1 - 3 years	Excel, St	Good salary	Bachelor's	Inability to	I haven't found	Yes, very chall	Communication	Finance/b	Not satisfied	Difficult	
28	27	Less than a year	Excel	Interested in	High school	Confidentia	Yes	Somewhat chal	Technical skills	Healthcare	Somewhat sa	Very easy	
29	28	1 - 3 years	SQL, Po	Interested in	Masters de	Securing a t	Yes	Somewhat chal	Technical skills	Retail/E-c	Very satisfie	Very difficult	
30	29	Less than a year	PowerBI	Interested in	Bachelor's	Getting real	I haven't found	Not challenging	Technical skills	Technolog	Not satisfied	Somewhat easy	
31	30	5+ years	Excel	Interested in	Bachelor's	Overcoming	Yes	Somewhat chal	Communication	Retail/E-c	Somewhat sa	Difficult	
32	31	3 - 5 years	Excel, P	Interested in	Bachelor's	Data will sh	Yes	Somewhat chal	Communication	Finance/b	Somewhat sa	Difficult	
33	32	1 - 3 years	Excel, P	Interested in	Bachelor's	Landing a g	Yes	Somewhat chal	Communication	Finance/b	Not satisfied	Difficult	
34	33	1 - 3 years	Excel, St	Interested in	Bachelor's	Data collect	Yes	Yes, very chall	Communication	Finance/b	Satisfied	Difficult	
35	34	1 - 3 years	PowerBI	Interested in	Bachelor's	Getting a jo	Yes	Somewhat chal	Communication	Healthcare	Not satisfied	Very difficult	
36	35	1 - 3 years	Excel	Opportunitie	High school	Managing v	I haven't found	Somewhat chal	Leadership and	Finance/b	Satisfied	Somewhat easy	
37	36	Less than a year	Excel, St	Interested in	Bachelor's	The biggest	I haven't found	Somewhat chal	Communication	Finance/b	Very satisfie	Difficult	

OVERVIEW OF THE ANALYSIS

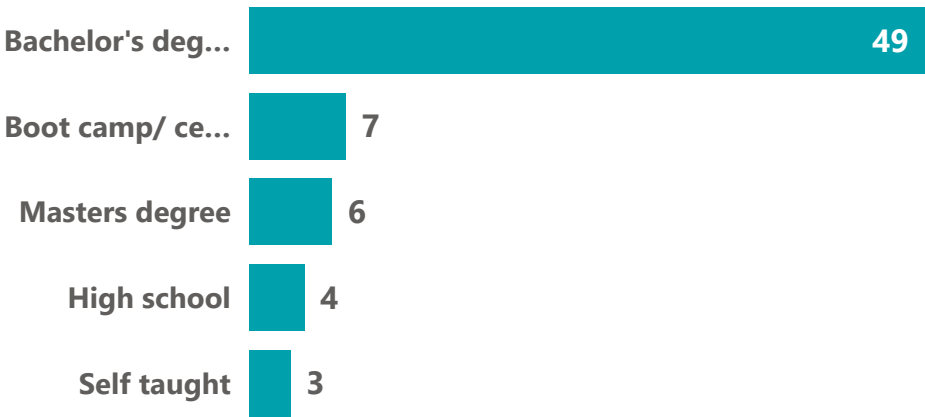
Insights from Analysts Demographics

- **Experience Distribution:** Entry-Level analysts represent the largest share of respondents. This finding is crucial for understand that the rate at which people are moving into the field is very high.
- **Education Distribution:** There is a huge representation of bachelor’s degree holder across the data this suggests that the people see data analysis as a job that requires strong intellectual capability.
- **How It Aids Decision-Making:**
Encourage people without tertiary education to come into the field by organizing events that can encourage them.



Experience L... ● Entry-L... ● Inter... ● Advanc... ● Specialist

Analyst by Education level



INSIGHTS FROM THE ANALYSIS

Insights from Tools popularity

- **Tools distribution:** **Excel 81%** and **PowerBI 65%** are the most used tool, **SQL 46%** and **Python 28%** come **3rd** and **4th**, with **R 14%** coming last

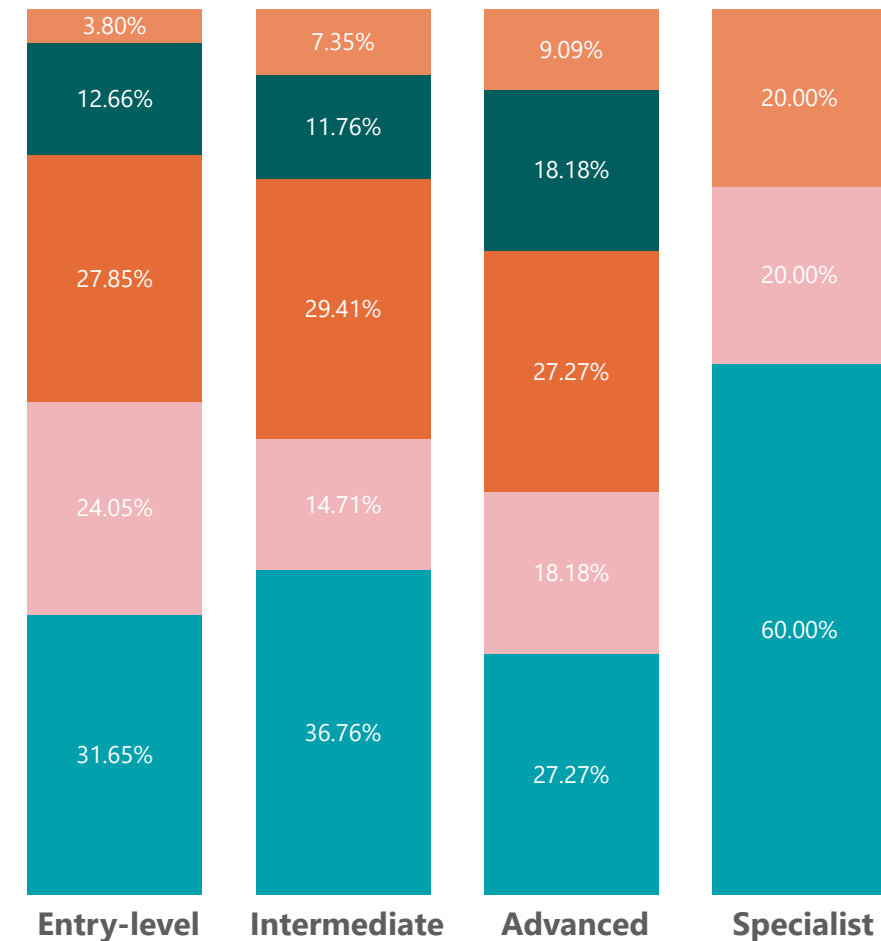
- **Education Distribution:** There is a huge representation of bachelor's degree holder across the data this suggests that the people see data analysis as a job that requires strong intellectual capability.

- **How It Aids Decision-Making:**

- Encourage people without tertiary education to come into the field by organizing events that can encourage them.

Tools by Experience

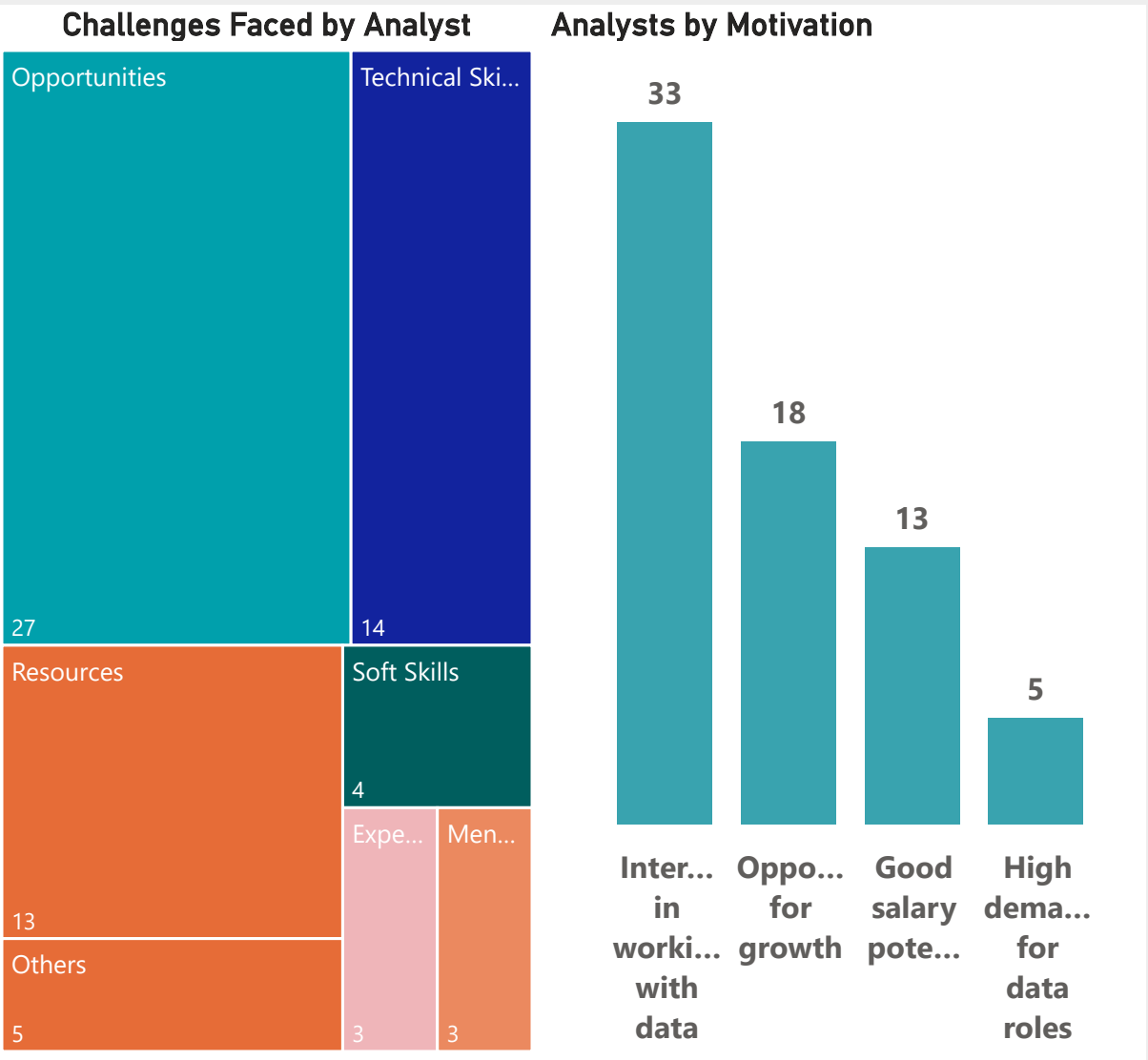
● Excel_ ● SQL ● PowerBI/Tableau_ ● Python_ ● R_



INSIGHTS FROM THE ANALYSIS

Insights from Career and Skills

- **Motivations for Data Field:** Majorly, people come into the field because of the interest in walking with data and the opportunities for growth, only 7% of the respondents claim they are in the field because of the high demand for data roles.
- **Top Challenges:** A large number of respondents face the challenge of finding opportunities like job roles and paid internships to advance their career. Lack of resources and the various Technical skills including coding stood out as to many
- **Satisfaction Level:** Only 6 respondents claim to be very satisfied with their careers in the data career, all these people use a minimum of 3 tools



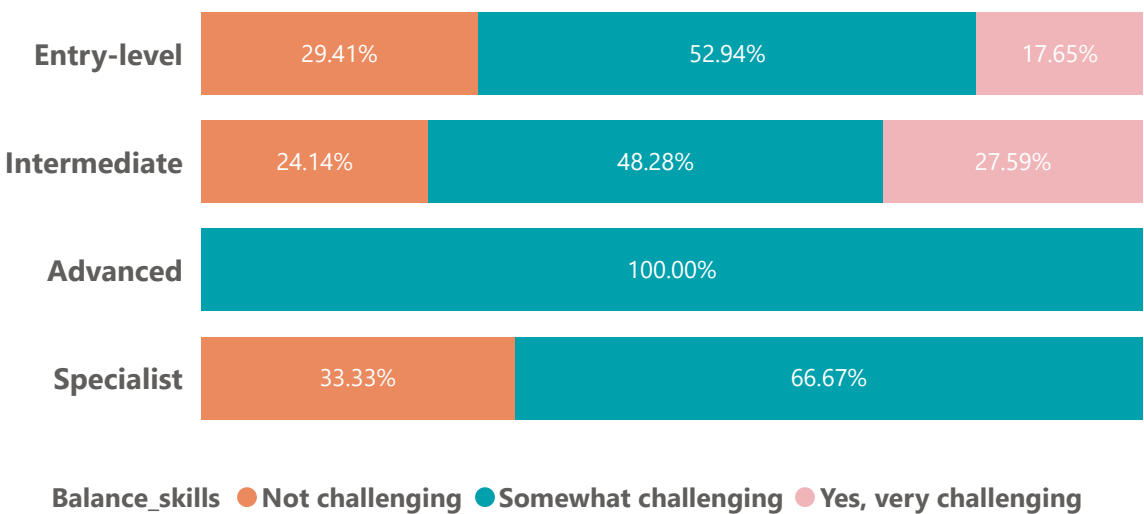
INSIGHTS FROM THE ANALYSIS

Insights from Skills and Learning

- **Top Valuable Skills:** About 70% of respondents believe that communication and story telling skill is the most valuable skill for advancing in a data career, Technical skills are deemed important too, but only very few people believe that Leadership and management skills are very important
- **Balance of Skills:** The highly experienced respondents balanced technical skills with soft skills like communication and presentation better than others with lower experience, some **Entry Level** and **Intermediate** respondents count it to be **very challenging**.

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Balancing Skills by Experience_level

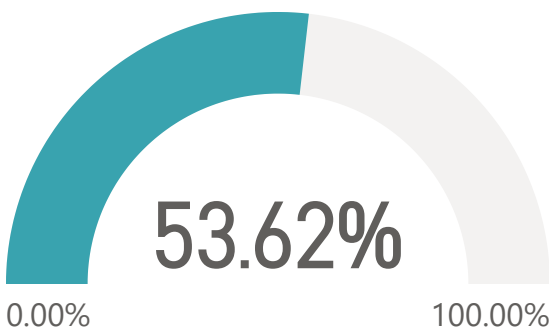


INSIGHTS FROM THE ANALYSIS

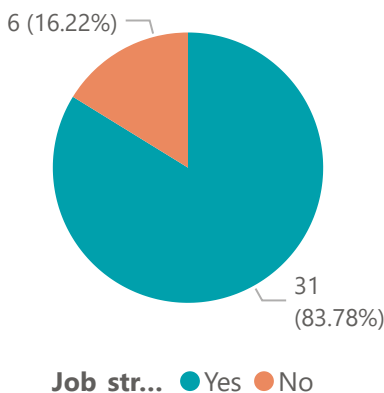
Insights from Tools Job Market

- **Employment Rate:** Only 53% of all respondents currently have a job, with entry-level respondents contributing 75% of the total unemployed respondents. Percentage of respondents who face challenges securing data-related jobs.
- **Job Struggles:** 84% of all employed respondents struggled to get a job in the data field. Respondents with higher education qualification (Bachelor's and Masters) however have a high tendency of getting jobs than others

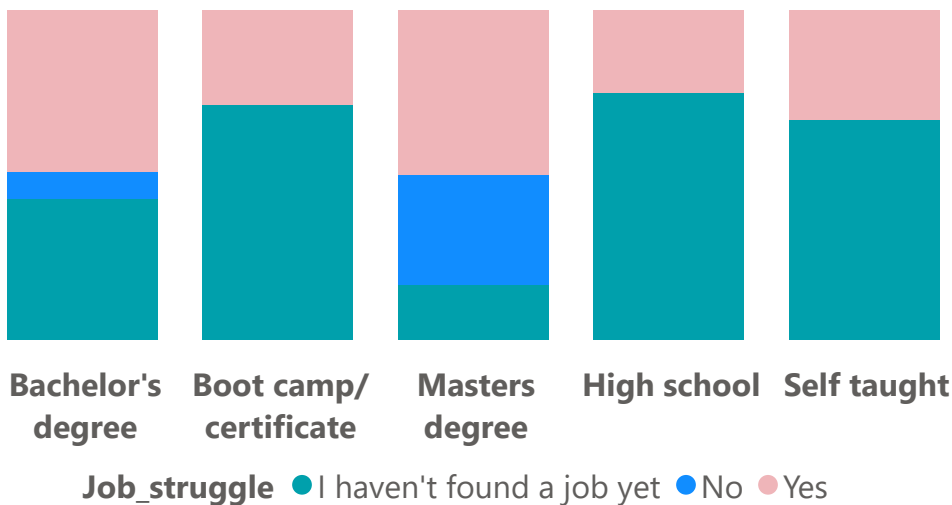
Total Working



Job Struggle by Analyst



Job Struggles by Education Level



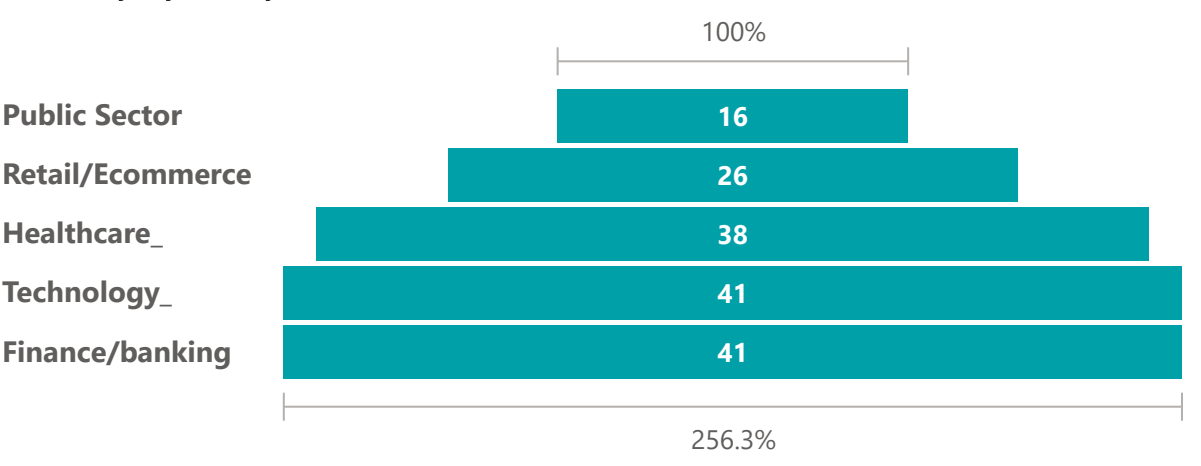
INSIGHTS FROM THE ANALYSIS

Insights from Job Market (Cont.d)..

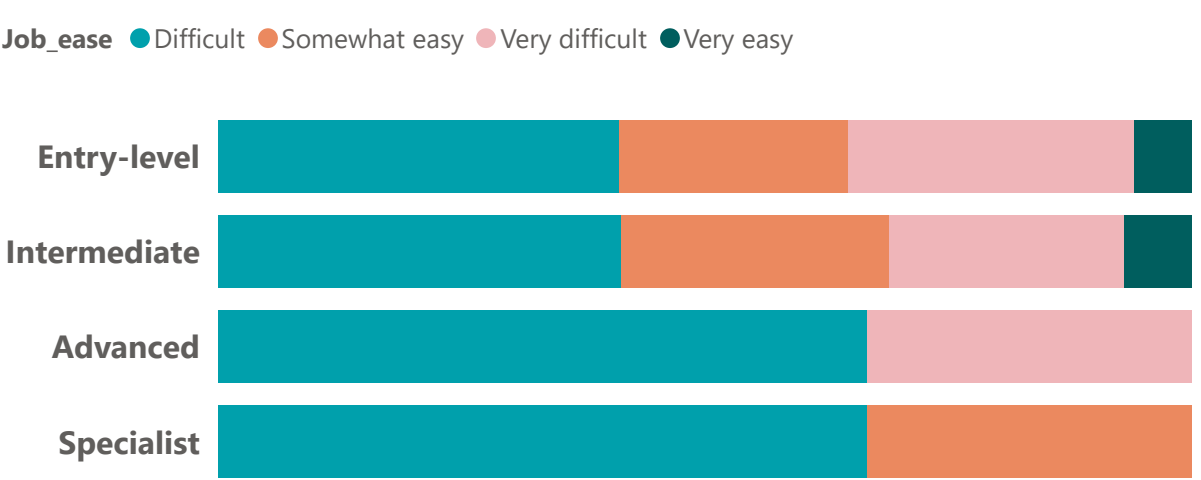
Industry Demand: Finance/banking and Technology are Joint best rated Industries (41) , Healthcare(38) and Retail/E-commerce(37) come as 2nd and 3rd with Government/Public (16) last

Job Ease: The respondents majorly found data roles to be difficult with no respect of education level, however, self taught respondents considered it very difficult to find data jobs

Industry by Analyst



Job Ease by Education



INSIGHTS FROM THE ANALYSIS

Are Participants Challenges Aligned with their Industries of Interest?

Participant Challenges & Industries : Participants challenges remains largely constant irrespective of the industry they look forward to working. as seen in the pivot table, opportunities, Technical Skills and Resources remain the challenges mostly faced across all industries where as all respondents also count soft skills and mentorship as the least challenges. Over 33% of respondents in each industry believe that opportunities in the most pressing challenge

Key_challenges	Finance/ banking	Technology_	Healthcare_	Retail/ Ecomm erce	Public Sector
		▼			
Opportunities	15	14	17	11	6
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Experience	2	3	2	1	1
Mentorship	3	2	0	1	1
Soft Skills	2	2	2	2	1
Others	1	1	2	3	1

INSIGHTS FROM THE ANALYSIS

Are Participants Challenges Aligned with their Skills of Interest?

Participant Challenges & Skill set : Respondents rated communication skills largely across all the industries . as seen in the pivot table, Communication and Story telling, and Technical Skills and remain the best rated skills across all industries where as all respondents also see Business acumen and Leadership and management as the least valuable skills. Over 62% of respondents in each industry believe that Communication and Story telling in the most rated for a data professional.

Valuable_skill	Finance/ banking	Technology_	Healthcare_	Retail/ Ecomm erce	Public Sector
		▼			
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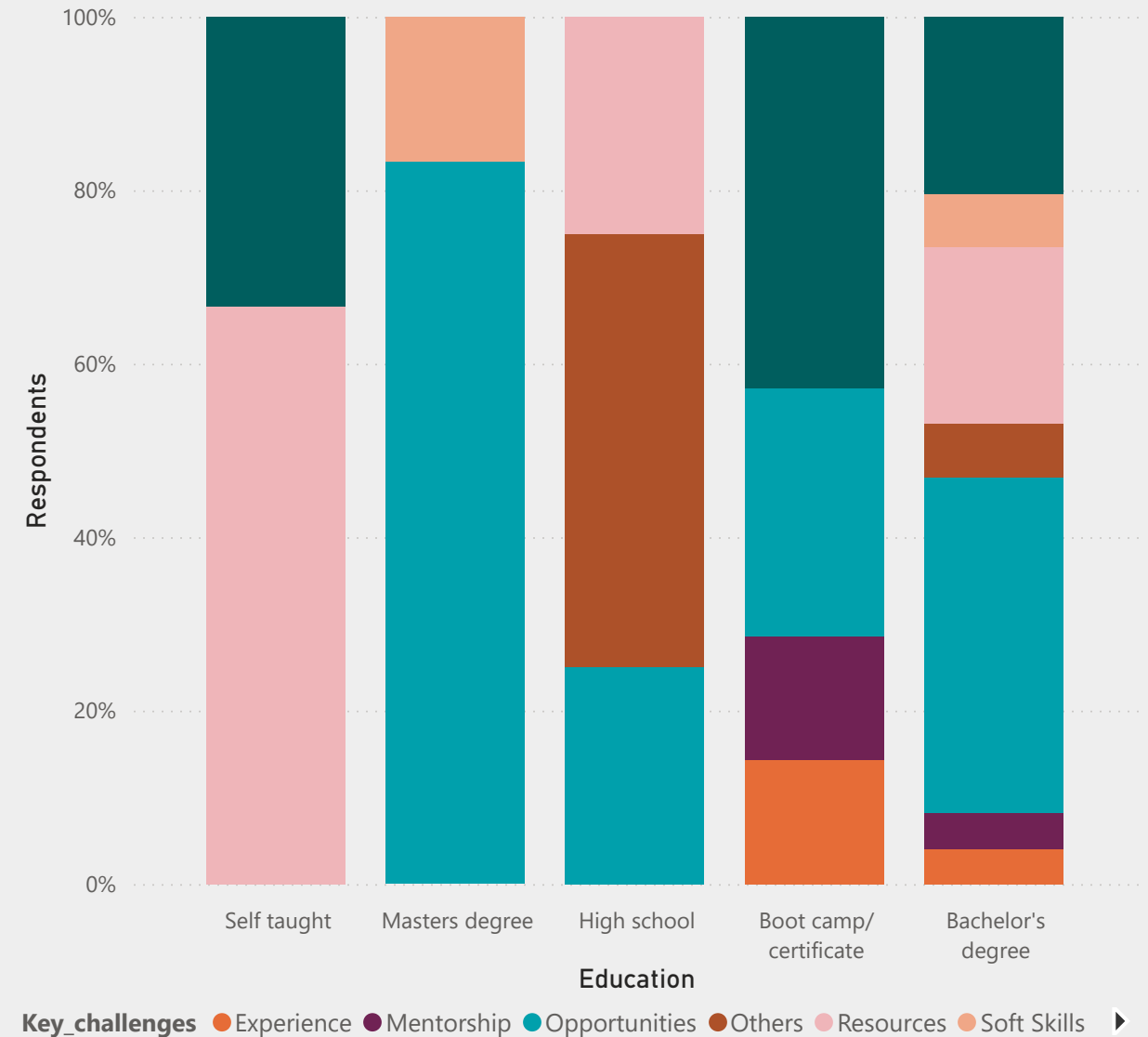
INSIGHTS FROM THE ANALYSIS

Are Participants Challenges Aligned with their Education Background?

Participant Challenges & Education Background :

Respondents with advanced education i.e Bachelor's or Masters face the most challenges in the area of opportunities, however others i.e Self taught, Bootcamp and Highschool face huge challenges with mastering the technical skills, may be as a result of their not being as intellectual as the other category.

Respondents by Education and Key_challenges



INSIGHTS FROM THE ANALYSIS

Are Participants Valuable Skills Aligned with their Education Background?

Valuable Skills & Education Background :

Respondents from all categories of Education level highlighted **Communication and Story telling skills** as the most important skill for a data professional, **Technical skills** come second as generally accepted with 33% of **Masters** degree holder . **28%** of **bootcamp** graduates also believe that **Buisness Acumen** is highly important.

Respondents by Education and Valuable_skill

